General questions/key insights

Q: Given this is a very large and conceptual course, myself and my peers are confused on what we actually need to study and know to do well in the exam.

A: Good question. You can have a good sense of the exam by doing the "Exam-like questions" in Learn for each module.

You don't need to know everything and you will have access to your notes and slides. The learning objectives can help you to focus, if you feel overwhelmed.

If you managed to finish the exercises for a lecture, you should be able to get 100% of the test.

Also, I should add that the "Exam-like questions" are harder than the actual exam. You can check the actual exam question in the exercise list for solar.

Q: I am not sure about how much of the content we need to know, especially on the slides showing technical images with much detail that are not fully explained.

I understand; you don't need to worry about knowing every detail of each image.

I tried to add a comment beside every image explaining what the main point is. If you understand that, you should be more than fine in the exam.

You can check the exam-like questions or activities to see how this is covered in the exam. If you understand the questions and answers, you will score well.

Q: I think some of the excercises confused me. Specifically the process you have to go through to find the right formulas and then changing them in the right way. I could definitely benefit from a step by step of how you generally solve a physics exercise. Like how to choose and change the formulas.

I understand. I'll try to spend some time next class going step by step.

My best advice is to do exercises; this way, you can develop an intuition on how and where to use an equation.

Q: I really enjoyed the engaging activities throughout the course — they made the material much more interactive and helped build intuition. That said, I found the exercises to be quite challenging relative to what was covered in class. A bit more alignment between the lecture content and the difficulty level of the problems would be helpful.

Thanks for the observation. The main purpose of the activities in class is to keep you "active" and "examplify" a concept.

To do so, it is more effective to use "simple" exercises or something a bit outside of the "exam", like the Geoguesser one.

In other words, if I cover exercises more close to the exam, it is likely that the students will feel more "tired" than "activated".

That being said, I understand your point, and I'll try to be creative on aligning the two better whenever possible.

Q: I liked the activities.

Thanks :) I'm glad

Q: The exercices are more complicated then the lecture, it is something hard to answer the quizzes.

Noted. I should mention that the "Exam-like questions" are actually harder than the exam. So you can also check the activity list to get a sense of the actual exams.

So, I hope the activities feel less hard. In any case, I have this in mind when designing the test, aligning the difficulty more with the lecture (instead of making the lecture harder).

Q: Understanding the way electricity is harnessed by a PV cell was more difficult to understand but was explained well. The class was very engaging though compared to others

I understand, it was also harder to explain. I didn't have as much time as for the first part to polish the slides and explanations. It is a work in progress. Thanks for the feedback.

Q: Having a formula booklet would be very helpful

Yes, and we do have one: Formulas for 47202 Introduction to Future Energy - ver02-12-2025

We may update something, but most will be the same.

Q: Everything is good so far

That is great! :)

Q: Hey! It would be nice if you could go through the formulas in more detail, explaining how and when to use them and what types of problems they can be used for. For example, you could first show us an easy problem and then give us an example of a more difficult one. It would also be really helpful if you could show us how we should think when approaching a problem. Even though you sometimes understand the teoretic behind it, it can still be hard to solve the problem if you never seen it before.

Thanks for the suggestion, I'll try to do more activities with formulas and explaining the solution to do some "scaffolding" for the students to be more prepare for the exercises.

Q: A key insight for me was understanding that power is not the same as energy. Power tells us how fast energy is transferred or used, which helped me understand problems involving time, such as filling a water tank or calculating energy consumption.

That is right! It is confusing sometimes, but once you get it, it helps a lot!.

Photovoltaics

Q: My muddy points, would be the PV graph explaining, efficiency, and the ratio between current, voltage and FF factor.

It makes sense; I was a bit tired on that point, so I could not explain it very well.

There is a lot of information on that graph and you don't need to know everything.

What is key is to know where you can find the main variables (e.g., shunt current, open circuit voltage, etc.) and how you can "visualize" the FF (i.e., as the ratio of the areas of ideal cell vs. real cell in operation).

The rest is more a "nice-to-know" knowledge and you don't need to worry much.

Q: What makes certain styles of solar power plants more or less effective?

That is a good question, but what does "effective" mean to you? If we are talking about "making money" or "producing more energy" it usually boils down to the location. Certain locations are simply better because they receive more energy in average. If you mean "energy efficiency" then it usually boils down to materials. Certain materials have better properties than others, which allow us to create better designs for the same external conditions.

It is much harder to improve on materials compared to location, so that is why there is a lot of "rush" into being the first to develop a solar park in a specific place.

Q: I didn't really understand bandgap and fill factor

Bandgap is the minimum energy difference between the valence and conduction bands in materials. You can understand it as the minimum energy you need to give an electron to become a "free charge". This energy can come as heat or a photon. For instance, if you heat up an insulator enough, it will eventually become a conductor.

Here is a nice youtube video, if it helps:
https://youtu.be/AF8d72mA41M?si=_3H4gsYd0rZ_HifL

Fill factor is a performance indicator for PV cells, it measures how close an IxV curve for a real PV is to the ideal case. A cell with low FF is "badly manufactured" as it has a lot of internal resistances, while a high FF cell is "better made". This not necessarily mean that it is more efficient! Because efficiency is also connected with bandgap and the light

spectrum. For instance, even you have the best PV cell made, the efficiency will be low if the bandgap is too low (because most energy will be converted to heat).

Q: Crystalline silicone cells and the PV caracteristic curve

Silicon cells are the most common PV cell type. The main information to retain is that they are made as a pn-junction with Boron and Phosphorus as the impurities to create charge donors and acceptors. Besides that there are some additional things you can do to avoid losses, which can arise from not collecting or separating the charges appropriately. They are listed in the slides and are "nice-to-know" things. Another good point to keep is that, although we can make more efficient cells, we stick to these ones because they are good enough and relatively cheap to make.

The PV characteristic curve holds a lot of information which you don't need to know all. The main point is to be able to locate some of the key variables/metrics in that graph, such as open-circuit voltage, shunt current and the fill factor. For the purpose of this course you can simple "take for granted" that the curve look they way it looks.

Q: How a semiconductor works and the logic behind band gap and doping

There is this great video from Veritassium, probably it can explain it much better than me and with animations: https://youtu.be/AF8d72mA41M?si=_3H4gsYd0rZ_HifL

I hope it helps.

Q: muddy points:
-ratio of solar capture area vs. collector area
-PV characteristic curve (Shunt specific current and Open Circuit Voltage)

1. The main point is that you can increase the efficiency of a solar thermal system by concentrating sunlight. This works because you can have a smaller surface area for the collector. The collector area is also emitting energy as heat losses, so reducing the size reduce the heat losses.

2. The main point of that part of the lecture is to understand the definition and where you can find it in the characteristic curve. So, for instance the shunt-specific current is the point where the voltage is null, while the open circuit voltage is when the current is zero.

Multiple topics

Q: - in formulas for absorbance, reflectance and transmittance, why do we denote rate of energy absorbed/ reflected/ transmitted by the body with E (reserved for energy) and not P for power, which is exactly about rate
- one of the ways to minimise heat losses in a solar thermal power is using materials with emissivities/ absorptances sensitive to wavelength of the solar radiation. I still don't really get why. why is very high emissivity and absorbance better for radiation of

shorter wavelength and very low emissivity and absorbance better for longer wavelength? how does the wavelength of absorbed radiation influence heat loses?
- in calculations for PV cells, why do we use current density j and not regular current I?
- what's the difference between between FF and efficiency of a PV cell?
- i am very confused about voltage and current and power in PV cell and dont even know what questions to ask (especially the diagram from slide 44 from the lecture on solar energy)

1. That was my particular choice. I was in doubt to use P or E. In the end I opted to "E" with a dot to avoid being confused with power in the electricity sense. But perhaps P makes more sense, I'll take note of that.

2. It is a good question. It is because the spectrum (i.e., wavelength distribution) of thermal radiation changes with the temperature. For low temperature, it is concentrated on long wavelenghts, while for high temperatures (like the ones derived from Sun) it is concentrated on short wavelengths (see Plank's law, https://en.wikipedia.org/wiki/Planck%27s_law). So, you increase the absorptance on low wavelengths to maximize absorption and minimize the emittance on long wavelengths to avoid heat losses. As this is a bit more than we should cover in the exam, I omited the explanation (which made it a bit senseless).

3. We can use both, it is just that current depends on the area of the PV cell, so it may be misleading to use current sometimes. For instance, you may think that a new cell is better than the old cell because it generates a higher current with the same voltage, but it is actually because the area is simply larger.

4. Fill factor is a metric of how close the cell is of the "perfect version" of itself, with no losses to resistances and etc. That does not necessarily means that a cell that has a higher FF will have a higher efficiency, because efficiency relies on the bandgap and the solar spectrum (i.e., distribution of wavelengths). For instance, I could make a perfect PV cell (FF=100%) with a material with very large bandgap (10 eV) and it will still be very inefficient because the sunlight photons can not form an electron-hole pair (i.e., efficiency = 0%).

5. Yes, it seems that it was confusing to most students because there was a lot of information there. The important thing to know is (1) "where" the variables shunt-current and open-circuit voltage are, (2) how you can visualize the fill factor as a ratio of areas, (3) that max. power is obtained in an intermediary current/voltage.

Others

Q: My muddy point is understanding why the COP equations for a refrigerator and a heat pump look almost identical, but use Qc and Qh differently in the numerator. I understand that the difference comes from what we consider the "useful" output, but I'm still confused about how to identify which heat transfer should be in the numerator when looking at a problem. Is there a simple way to think about the system perspective so I don't have to memorize which equation goes with which device?

I'll try my best to explain how I "memorize it". Basically:

$$\eta = \frac{what\ you\ get\ of\ useful}{what\ you\ pay}$$

The top and bottom changes depending on the type of thermal machine

Machine	What you pay	What you get of useful
Refrigerator	Power = W	Heat transfered from the cold source = Q_c
Heat pump	Power = W	Heat transfered to the hot source = Q_h

Now from the first law, we have:

Energy in = Energy out

$$W = Q_h - Q_c$$

*: Here power is positive

Then, we can write the efficiencies as:

$$\eta_{refrig.} = \frac{Q_c}{W} = \frac{Q_c}{Q_h - Q_c} = \frac{T_c}{T_h - T_c}$$
$$\eta_{hp.} = \frac{Q_h}{W} = \frac{Q_h}{Q_h - Q_c} = \frac{T_h}{T_h - T_c}$$

The reason we can jump from heats to temperature it is "because of the second law" for a reversible process:

$$Q = T \cdot \Delta S$$

And it "just so happens" that DeltaS is the same for both heats.

I hope that helps, I took somethings for granted some things in this explanation because going too much in deep would exceed the Bachelor level.

And remember that this relationship it is valid only for the ideal cycles!